

SceneSenseAI Repository Guide

Folder Responsibilities and Usage

app/ - System orchestration and pipeline logic.
models/ - AI model wrappers (Detection, Depth, Flow, etc.).
configs/ - YAML configuration files.
checkpoints/ - Model weights and checkpoints.
datasets/ - Raw, processed and annotated datasets.
notebooks/ - Experimental notebooks.
experiments/ - Weekly research logs and results.
evaluation/ - Metrics computation scripts.
outputs/ - Generated outputs (videos, images, JSON, audio).
logs/ - Runtime logs.
utils/ - Helper utilities.
tests/ - Unit tests.
scripts/ - Entry-point scripts.
docs/ - Documentation and design assets.
paper/ - Research paper material.

Key App Files

pipeline.py - Main execution loop
scene_builder.py - Creates unified scene dictionary
scene_schema.py - Defines scene structure
narrator.py - Wearable narrator mode
sentinel.py - Surveillance mode
vlm_router.py - Selects VLM backend
graph_reasoner.py - Scene graph reasoning

Current Development Backbone

Webcam → Detector → Depth → Flow → Scene Builder → Scene Dictionary → outputs/json